

Project Initialization and Planning Phase

Date	01 July 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	Develop an AI and Machine Learning based agricultural production optimization system that recommends the most suitable crop using N, P, K, pH, temperature, humidity and rainfall.
Scope	Provide crop recommendation, crop suitability analysis, environmental assessment, and research insights through a web application.
Problem Statement	
Description	Farmers often select crops without data-driven guidance, leading to low yield, poor resource ut
Impact	Improves productivity, optimizes fertilizer and water usage, increases farmer income, and supports sustainable agriculture.
Proposed Solution	
Approach	Collect soil and environmental parameters, preprocess the data, train a Machine Learning model, and recommend the most suitable crop through a Flask-based web application.
Key Features	• Smart Crop Recommendation • Crop Suitability Analysis • Environmental Assessment • AI/ML Prediction Model • Farmer-Friendly Dashboard • Research & Policy Insights

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 Processor
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Python frameworks	FastAPI, Streamlit
Libraries	Additional libraries	Pandas NumPy Scikit-learn Matplotlib Joblib FastAPI
Development Environment	IDE	Visual Studio Code, Jupyter Notebook, Git & GitHub
Data		
Data	Dataset	Crop recommendation dataset in CSV format containing Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, Rainfall, and Crop Label. Crop Recommendation Dataset (Kaggle - CSV)